

ActInf Textbook Group ~ Cohort 8 ~ Session 18

(Chapter 9 & 10) ~ 10/02/2025

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Okay, greetings. So, this is cohort 8 in the discussion on chapter 9 and 10. They're good chapters. They bring together a lot of questions. 10 is of course an overview and looking out in the end of the book and nine deals with a lot of experimental topics. So, does anyone want to start with a section or a quote or a question from what they've been working on in the time between or anything pre-existing in the document or anywhere to start? It could be on nine or 10 or as 10 sort of deals with everything. It could kind of be any any any topic that anyone wants to bring up. existing section that someone highlighted or a question they wrote down or or we can just go through nine Okay, just raise your hand in in Zoom or write in the chat if you have a specific thought or if you want to go somewhere else. So nine is going to flip the direction that the generative models are being deployed in relative to the rest of the book. In the chapters leading up to nine pretty much ad hoc or ab initio the from starting position a generative model is specified and then it's used to generate synthetic outcomes. Nine is going to flip that by saying okay what if we already have empirical data but we want to recover or infer parameters that are unobserved from empirical data. That's what this chapter is all about. going from having a generative model of a situation kind of like a music box like you built it, you run the crank and it puts out music. Well, now there's a situation where you have a recording of bird song or mouse behavior or whatever it is and you want to go back to recover the structure or the parameters of a generative model. So this is a more empirically grounded. This is the kind of stuff in the figures that you'll see like in a paper Michael or Okay. just anyone can unmute if they want to ask a question. Um, and and as always with this group of authors in the book, there's just so many fun standalone spicy quotations that that can be discussed even with any background. 9.2 two goes into a very important structural perspective on cognitive modeling on behavioral modeling and it's represented in this figure 9.1. So now we're looking at that kind of figure 4.3 figure 7.3 discrete time generative model. So that that figure would have looked just like that. But now we're saying okay so that's the artifact that

we're making is the pom dp that has this graphical structure but let's think about our laboratory as well. What we're calling observations for the mouse or for the agent who we're modeling are under the control of the laboratory apparatus and then what we're calling actions or outputs of the agent that we're studying are actually observations for us as researchers. And there are parameters of our laboratory that set up the context for the situation. So it's a good reminder that just because something is called like O or observation, it doesn't mean that it is simply only an observation. It's that's a perspective on the direction that that data point is used for. And here's kind of zooming out to the um the bird in the laboratory and a reminder that once an observation for the system of interest could be under the control of the experimentter and that is going to come into play because now we're thinking about like from the research lab or behavioral studies how do we talk about that generative model. Okay. Um there's two reasons for fitting a computational model to observe behavior or you can come up with more lump or split however you see fit. This is what they're going to focus on. This is the rationale for chapter 9. The first is to estimate parameters of interest from that model that best explain the behavior of specific subject or group of subjects. So simple example, if we're going to flip a coin, we're going to get this series of heads and tails, but there's this unobserved variable that's nowhere in the natural empirical world of like what is its probability of landing on one side or the other. That's not a feature of the world that can just be measured with a stick, but it might be a latent factor that we're very interested in, like is this a fair coin? So there's unobservables in the world that are whether they're mechanistically related to something that exists but we just didn't measure or whether they are something that that doesn't specifically exist in any location. There's all kinds of latent states in the world that we want to infer. And big picture that basically is what makes something cognitive or computational phenotyping. The second reason is to compare alternative hypotheses. So different model structures that represent different explanations for behavioral phenomena. So in the first situation, we're conditioning upon a structure of a model and then we're going to run it back to see if individuals or treatment groups differ in some aspect of that model. Like we have a latent model of suitability uh and then we're going to see if two birds differ in how they're treating suitability. The second situation is like we have a bird movement data set and we have two different models that might explain suitability and now we're going to compare which of these models of suitability is more parsimonious or explains the data better. So in this empirical grounding sometimes we're just going to have one model and we're going to run it back to see how individuals and groups differ. A second situation is we're going to have a portfolio of models

and we're going to go from the empirical data to ask which of these empirical uh which one of the generative model architectures have the most empirical support. Um 9.3 is kind of a mathematical aside on variational applause. This is a way to fit distributions with a central tendency. There's more detail we can go into, but big picture, it's a incrementally optimizable strategy for fitting a central tendency to a distribution. 9.4 poor parametric empirical bay or peeb sounds kind of poly salabic but it accords with a pretty intuitive way that people think about basian statistics which is there's always this question of well what about the prior or isn't there an influence of the prior on the posterior there is that's kind of the whole point so how do you ever really know that the posterior results which are after data are not being unduly enclosed by your priors. So there's different strategies on this. One of them is to start with multiple different prior starting points. So we're going to have that coin and then we're going to start from a prior of a uniform prior kind of flat across all the 0ero to one and then we'll start with different starting points for more opinionated priors. And if all of those are converging on the same value, then we feel more grounded in saying that we're getting a posterior that represents the informational signal of the data. We're not just passing through our priors. Another way to have an empirical grounding for prior selection is empirical base, which is you have some data sets. So we have flipped the coin 100 times and it was 7030 heads to tails. So in that situation, we can just start our prior with 7030. So it's kind of like the first phase is a frequentist phase or just using a maximum likelihood solution. You're just taking the doing the classical frequentist thing, taking the plain maximum likelihood straightforward empirical results that you have for the first part of your experiment and then re-entering that with uncertainty into a basian model. So, let's just say you don't necessarily know um what should the blood dilation parameter be for this or that treatment group in this fMRI experiment. It's like it's not going to be something you can just figure out pencil and paper or with a computer, but you might be able to grab a subset or a different fMRI experiment and then say, okay, well, the range that we ever observed for that parameter was one to five. So let's just set our parameter as 0.5 to 10 to go beyond what has ever been empirically observed and then we'll sharpen it from there. So that's a short section just describing how and and this has gone into a lot more detail in the SPM textbook and and methods. This is basically just saying if you're concerned about prior influence on posterior as as basians are then you can do different strategies you can try different priors different precisions on priors or you can just start with setting your prior based upon what's empirically already found on a subset of the data. Section 9.5. This is kind of a lot like chapter six. That's the kind of Jimmyi Hendris if six were nine

part of this book, which is chapter six was the recipe for creating that generative model and going step by step in that forward direction. That was the recipe for building the music box to get to the sol. Now we're going to work back and say what are the steps if we have this data set and we are trying to do analysis on the data set in order to get to a publication or something that looks like that. So here are the steps and here's the SPM package. It'll be kind of a um a little bit of an exploration away from the textbook, but we could look at SPM if people want or perhaps better in a separate group. So, first collect the behavioral data. There's a ton that could be said about ethology and behavioral data analysis, but this is going to look like what's the real-time data that you're describing? How' you how did those behaviors get defined? What are the parameters that you're observing? You know, what's the temperature? What time of day was it? What did you define this behavior as? How did you measure it? Okay. Step two is chapter six. Here they say chapter 7 because they're talking about the discrete time VP, but maybe they meant chapter six actually. Um, more generally, right, you do need to build the generative model that's going to receive and and output the structure of the data that you're looking for. That's where the generative model that you write is going to come up alongside and recapitulate the empirical structure of like the stimuli that you controlled in your lab. You want to have the the observations incoming to the rodents. The actions that you're studying the output of the rodents, you want to be the behaviors that you observe in the lab. That's the kind of closed loop. Specify a likelihood function. You can look back to chapter 2 with like the likelihood of the apple or the frog jumping. Step four, specify the prior beliefs. That was this earlier discussion. You could start from a variety of prior positions or you could do the parametric empirical base use some standard inference scheme like variational applause. So that's why it was brought up earlier in this chapter to kind of go back from the observations and then ask well what's the most likely parameters of the underlying model that would have given rise to those empirical observations and then do ANOVA like statistics. So analysis of variance, this is where you start getting p values and things like that for looking at how different groups and within and among groups or within and among treatments have variability in those latent inferred parameters. Because if you just wanted to say uh do an ANOVA on the descriptive statistics, look at just primary empirical variation within and between groups in measurements. You could do that with a purely descriptive framework, purely frequentist framework because you're not inferring any latent variables. There's nothing that isn't just being empirically set from the measurement. But if you want to start dealing with features of the situation that aren't directly measurable, like again the coin tendency is not

directly measurable. So if you want to talk about comparing the coins in terms of their propensity to one side or the other, you're talking about variables that aren't simply observable. But if you just wanted to flip a bunch of coins a thousand times and then just say, well, which one came up heads more? You don't even need to do statistics. So here's what those steps look like and this is in reference to chapter 7. So we already had this data set. Maybe the previous grad student in this team a setting gave us this data set. The generative model Allah chapter 6 and 7 gets formed again just like figure 9.2 to kind of closing the loop. Other side of the coin, other side of the handshake from this rat. Chapter 7 was looking at it from the rat's perspective. Now we're looking at it from the laboratories perspective in terms of what data we're going to analyze. priors and likelihoods come in to this model inversion step which can be a little tricky to get set up but there are standard methods and there's some examples in SPM and other toolkits but then what you end up getting are projecting from observations of data and then basically the empirical data the likelihood and the prior come together like in this reactor or crucible of the generative and then it projects onto this latent variable which could be like um hunger level. So maybe we find that as we we suspect that as a function of time since eating the last food that the the rodent is going to differ in its decision- making and um we could test the hypothesis whether being um more hungry is going to lead to it being more um risk-taking and just going for the the gamble or whether it's going to improve its its seriousness and that's going to make it go to the epistemic queue more. That would be like kind of end level hypotheses that we want to have empirical evidence for. And we say, okay, we did five different treatment groups. We had five different intervals of of timing after last eating. And then we inferred this parameter. And here's the mean and the variance of each of those replicates plotted as a bar chart. And then that's where we can do an ANOVA. You can say these three, you know, do or don't significantly differ from these two. These two do do or don't significantly differ from each other. And then that's kind of the result would be like, you know, early phase this, you know, but not, you know, or late late phase hunger but not early phase hunger increases exploratory behavior in a rodent. And that's kind of the paperlike conclusion that gets yielded at the end of all of the steps. But the important feature was we're talking about differences within and among groups that are not simply a descriptive feature of the world. So we have to do this kind of latent or abducted or inferred variable. And we want that abducted variable not just to be fantastical but something that has a structurally typed and interpretable basis where there's modular prior and likelihood specification. So we can come to a empirically grounded scientific conclusion by this stage. 9.6 goes into some settings

where that type of process has been followed. Um this is two different eye movement examples. The left one is in a continuous setting and the right one is in a discrete set. 9.7 goes a little bit into this kind of territory of false inference, but of course it's a real inference. It's just not necessarily of a radical conclusion. This is sort of like there there's there's a ton of of of harder and softer discussion more on just the linguistics and just what's a pathology, what's normal, all those kinds of things all the way on through the the real details of parameterizing cognitive models. But table 9.1 has a variety of these different phenomena and citations to explore how they're dealt with. This is a good um pointer paragraph, but you know, why does this matter? Why can't we just look at the empirical data? Why can't we just look look at people's behavior and just classify them on on simple descriptives? Like what if we have a model like this, we could start to explore what scenarios or what data sets would we want to collect to determine like okay we're seeing this kind of behavior you know insert your favorite phenomena. Is that due to the prior beliefs being a certain way? Is that due to there being a inability to update latent states? Is that due to uh offkilter belief about how latent states observe uh connect with observables? So those are the kinds of of interpretability that you get with a phase graph of cognitive modeling. You could also just put a big neural network or LLM transformer and that could do a great job with classification, but then you might have low interpretability. And so that's the kind of funny thing is then you need to do this whole second layer of interpretability analysis to look at like I vectors of activation in the neural network to get back down to what you hope might be a semantic or a symbolic aspect of behavior. the one approach isn't isn't uh better. You'll see many papers arguably vastly more that just take um single or multimodal data set or gene expression or neuro imaging and just do machine learning or just we did a neural network classifier to classify cancer versus not cancer. It's it's not a mechanistic or symbolic account underlying cancer. It's just data processing and clustering. So that's where the main contents of the book get the reader. Penn is going to revisit and extend on a on a review footing and and open up some some ongoing directions and kind of leave for for a consideration and discussion about is active inference a unified theory of sentient behavior. But in terms of the the delivery of the upskilling of this book, it's really the two through nine chapters where chapter 1 and 10 are those bigger picture review. Chapter 2 and three got us to chapter 4 with generative models through the low and high road. Chapter five reviewed the mammal neuroscience papers using generative models in mamleian neuroscience. Chapter six was a recipe for building a generative model. Chapter seven and eight went into generative models in more detail in the discrete and the continuous

time setting. Chapter nine talks about running it back and going from empirical data to parameters that you can infer about your generative model and then the kinds of papers you can write with that. Any any like random thoughts or ideas or questions on not Um, in chapter nine, I'm still kind of intrigued or uh stuck on the observations being related to the experimentter and um I think figure 91 Yeah. >> So, let's just say that um the generative model of of the the mouse and the T-maze and there's going to be like two different colors flashed of light and it's going to have to make a decision in the T-maze based upon two different flashed colors of light. or there's an observation for the rodent that's underneath the bottom position of the teammates like the epistemic cue. So there's stuff like the agent that we're studying, the rodent in the lab or the bird outside or the person in the clinic or the LLM that we're studying in the sandbox or whatever it is, there's something that's going to be an inbound data point for it, that flash of light. But from our laboratory setting, we're controlling the sequence of the flash of lights. So it's an action from the lab, but it's an observation from the perspective of the agent who's integrating it. And then analogously, the action selection like expected free energy, policy inference, all the stuff we're talking about with action selection results in a movement action. But that movement action is an observation for us recording its data. So that kind of takes us back to the markoff blanket where it's like the O and the U are the the two are are the two directions on the blanket interface. And so whether we think about that flash of light coming from us as an action, it is also going to the rodent as an observation. And then the action is getting emitted from the rodent as an action, but it's getting received by the observer as a data point. But it's just two different views on the same interface. and just very, you know, interesting with a dash line here and a dash line here. So that's kind of like where the the hyper priors kind of at some point you just say, okay, it's just that's where the hyper prior stop because you could you okay, but how did we choose which light sequence to flash? We can look at that policy inference. But just it the great part about basian inference is it makes where makes what your priors and what the structure of your problem. It makes it clear what you did and didn't leave open as a degree of freedom versus just like well here's the data we collected. The data speak for themselves. Okay. Pen is going to review and connect dots. Here's possibly one of the contentious and and discussionoriented claims. Does active inference provide a complete solution to the adaptive problems that sentient organisms have to solve? So, a lot of ways to to take that, right? Is there any part of 10 that somebody wants to go to or or a question that they want to explore or should we just also continue with this linearly? So, 10 two they're going to summarize the earlier chapters. Chapter 1, chapter 2, chapter 3, chapter 4,

chapter 5, chapter 6, 78, 9. more words but similar to what I said earlier connecting the dots here they introduce Daniel Dennett's concept of the whole iguana which was like a thought experiment saying hey maybe instead of going deeper into one specific cognitive function or into multiple specific cognitive functions and then thinking that we're going to figure out the lateral the lateral connections later. How about start with a complete cognitive modeling approach for something like an iguana although I will say that insect whole body and brain modeling is way more advanced and neat as well. again more discussible slashcontentious claims for people who who who could take pro and con and and and differ in a variety of ways. One benefit of active inference is that it offers a first principle account of the ways in which organisms solve their adaptive problems. Like what is a first principle account? What would be a first principles account of how organisms solve their adaptive problems? Does that already exist? Could it exist? Could there be multiple first principal accounts? Are we focused only on how organisms solve adaptive problems? Is that kind of the problem to solve? The normative approach pursued in this book assumes it is possible to start from the principle of variational free energy minimization and derive implications about specific cognitive processes. So this is what it means to have a unifying cognitive model. That's what the chapter, that's what the book, arguably, that's what the field is looking to deliver on is instead of this portfolio of specialized models with different architecture, different notation, different ontology, different software packages, different research communities, etc., etc., etc. for memory, anticipation, and all these different cognitive phenomena. Maybe perception, action selection, attention, emotion regulation, etc. could be addressable within a unified approach to cognitive modeling. They kind of walk through that how this um softly cybernetic autopoietic but ultimately just kind of organismal realist perspective helps us integrate cognitive phenomena. By by keeping in mind that there's actually a creature out there having to do stuff in a niche, we are going to get to the point of being able to consider how memory, attention, planning, etc. come together. It's kind of like the nothing in biology makes sense except in light of evolution question. This is kind of like saying well nothing in cognitive is going to make sense except in light of the cognitive scenario that is being discussed that could be seen as taological but it also is actually the the the space where all these otherwise fragmented phenomena make sense. different ways that pre that that previous taxonomies of cognitive functions have have been applied. Th those taxonomies may or may not be aligned with different levels of kind of Mars analysis. So there's just there's an explosion yesterday and today and tomorrow of cognitive ontologies and so what the scope of coverage of those and

the way they overlap and everything there's just a lot of ways to slice this cake. They're going to argue that having this kind of a normative perspective, which makes it sound like it's the socially normative perspective, but but there's other things that could be kind of inserted in there, like adopting a generic unified system, scientifically based ontology with useful educational materials and implementable open source software may also help in identifying formal analogies studied in different domains. Like if we want to talk about memory in LLMs or if we want to talk about anticipation or planning of groups of cells or something, are we just going to leave that as a linguistic metaphor or simile and just say that it's kind of it's useful for communication, but it's not going to help us actually move that field of research forward? Or could we go beyond just remarking on similarity and actually talk about some of the the real isomorphisms and even cross apply different models from different systems. So we get to understand and and juxtapose and integrate a variety of phenomena within a given system like rodent. But then also by having a unifying normative approach, we could also unify across different systems. And then it it sets up the possibility of developing more specific process theories that connect abstract symbolic cognitive processes to neuronal or cellular mechanisms. So we could have a um sensemaking or classifying agent. It could be an incilico computer agent. It could be a robot. It could be a mouse. It could be a person. They could all be solving or doing the same cognitive problem. And then we could connect that to specific process mechanistic theories like being able to do generalization could be mapped onto the columnar architecture of the neocortex. That's like the thousand brains idea. 10 four goes into prediction. There's more in common than different between predictive processing and active inference. But there's also important points of difference. If what is being predicted includes not just what is perceived in and in the niche in the world, but also what is to be done and then that prediction of action becomes enacted. That's basically active inference. there's a kind of you don't want to just continue to revive straw people arguments again and again and again, but there is this inertial gestalt that we're dealing with an influx of information. We're processing and sifting and fine-tuning the incoming information to come to a clear picture of all of this messy overload of data. That's like an inbound processing model. And the kind of different posture that understanding predictive processing and active inference puts you on is like oh actually the information coming in just visually for example it's sparse variable resolution variable amounts of color big blind spot etc. We're not dealing with a overload of incoming information and doing a sorting exercise. We're generating an unrolling visual model and it's getting just lightly updated through incoming data which is quite sparse. That connects

with this self-evidencing like understanding the imperative not to be to maximize reward on some auxiliary reward function but for things to have a firstness of just being what they are and then through their activity seeking to become what they are in this way that's both like very tlogical and deflationary and simple but also very esoteric 1041 goes more into predictive processing. So Andy Clark and Yaka Powi are some of the big authors in that area and active imprints builds on that with more formalism the high road free energy principle and software implementations. Whereas most of the discussion about predictive processing, it's either kind of qualitative philosophical like about the difference between that inbound versus more generative concept of perception um or it's specifically in reference to like specific neurocircuitry of the mammal brain. So kind of already at the level of a process theory perception. Kristen and others often go back to Helmholds uh and and the perception as an unconscious infrance. And then I always come back to um live stream 43.0 where Maria kind of took it back to like much that was a great presentation within this. Zero. So that that that helps just give this it again is easy to fall into that inbound processing feel and that is the trick basian brain pretty similar to um predictive processing in that it stays focused on the brain whereas of course the scope of active inference doesn't have to be brain centric action earlier section brought up perception. So it kind of the um passive predictive processing passive basian brain just is consistent with like the brain is like this observer individual behind a one-way mirror sense making and then is that sense making more about processing incoming information or is that about having a rich world model that gets fine-tuned with incoming data and then whether you come from predictive processing or basing brain or just go to active imprints. You also are like, okay, now that observer is going to be doing stuff. They're going to pivot in their chair. They're going to do is cades. They're going to move their head. They're going to flip pages in books. There's been a lot of discussions about how the kind of embodiment of primary sensory information becomes integrated with skilled action and more cognitive and symbolic kind of higher order cognitive features. have you go from just having joint control to spelling out messages in sign language sign language uh cybernetics uh they field there's a v variety of ways that people have looked at the perception action cycle like t this is kind of a a standard industrial engineering frame test operate test exit there's also the UDA, observe, orient, decide, act, and a variety of other ways that people have cut that cognitive cake. Kind of to the earlier point about there being such a variety of cognitive ontologies and just so many different ways. And that's why it helps to to be familiar with a bunch but be fluids in a few. Perception is signal processing and sense making. Action

selection is often called control. That doesn't mean total control. It doesn't mean lock step. It just means basically action re-entry into a situation. So sensemaking and control, perception and action, joint distribution, that's the whole iguana. So instead of just having the iguana locked into an apparatus and then study its discrimination behavior and then put it out in a different experiment and study its choice behavior as if it was getting perfect information from the environment. The whole iguana is let's model the perception cognition action of an agent as a joint distribution and then that can be a basian joint distribution and then we can do all these different equations in software um utility and and there's so many ways that people have gone with economics but utility as we kind of found from equation 26 utility is the pragmatic only special case of decision-m complimentary to the novelty only epistemic value. Whereas expected free energy is a synthetic scoring for policy selection that could be zero 100 could be 100 could be 80/20 could be 50/50. But in economics those kinds of discussions micro and macroeconomics those are classic discussions like are people maximizing reward or they satisfying basing decision theory example of how like just searching in quotations active inference you'll find some literature but I mean there might be a basian decision theory paper using an evidence lower bound using a variational free energy so by all means active inference and under the free energy principle they just might not use those words so it's kind of about reading between lines and um not just having syntax flags because there's relevant work that just uses a different or or work that's not written in English, of course, which is why it's important to have a core ontology that isn't just grounded in in a connotation of a term. reinforcement learning, tons of connections with active inference and machine learning, especially nowadays. Planning. Planning is action selection, including counterfactual futures. You quickly get into all the kinds of things that the chess algorithms were getting into in the 1980s like combinatoric explosion of policies and highly conditional sequences of action, stuff like that. Behavior general topic, right? as many. It just behavior is what is what things do. That makes me think about a paper of my PhD adviser from the plant behavior. Think about nowadays just podcasts and discourse and just how there's so much focus on things being on on things behaving and it's like those debates have are long standing just who gets to behave what's computation what's cognition all that bounded rationality a lot of topics here especially like the kind of AI alignment and safety space trivially all implemented systems are going to be boundedly rational and all systems are going to have this sense in which like if you account for what they are they're as rational as they are that's like the the what's optimal about optimal basian inference how can we say something pathological is optimal but that's the

whole part from chapter nine veance emotion motivation Some of these higher order psychological constructs, some of them even experienced that that people spend a lot of effort on on modeling. Homeostasis, allostasis, interception. Some more foundational organismal capacities that also can be modeled within this whole iguana approach. How how do we extend the um simple first order cybernetics like a homeostat into what what some sometimes people call second order cybernetics like a learning homeostat and then you get into like third order. But, you know, at that point, it's just like are we trying to to um you know what what linguistic ontology is just being extended out? What's the real question that's being addressed other than just pointing and and and labeling natural phenomena? more phenomena open directions. I guess this is where we get to learn and explore. These are definitely things you'll see in the discord in in the discourse in papers today. multi-agent social interactions amongst uh humans or amongst all AIs or augmented groups, robotics. Summary: Lord of the Rings, possible illusion to the ring on the cover of the book. Um kind of a personal like ending section of the book in terms of just the style of the writing and they lay it out pretty clearly. What what did they ask? What did they respond with? What was their evaluation? So, kind of a little bit of a Hegelian triad with these three sentences. They hoped and in the textbook group we hope to extend the introduction to those who use these methods conceptually and formally. Also the kind of third cousin there is the computational because the conceptual is often taken to me qualitative philosophical and formal to mean analytical mathematical and then it's easy to feel and actually even have high familiarity with philosophical and mathematical but still not to be able to even write a script that does a simple thing. So the computational is a really important piece too. And then there's this funny turn where they they kind of are like remember how everything we said about now go on. And then it might look exactly the same or it might be different. And then there's the appendences. So pretty pretty long chapter. I mean, you could draw you could draw lines coming out of this, especially some of these short like six paragraph sections. You could expand on it a lot. Oh yeah, the last minutes. Where does that kind of take anyone of any thoughts or questions on This part about the important rights of passage I kind of get and like theoretical neurobiology like Kristen's research group is called the theoretical neurobiology group. So in this part I kind of got the sense of Par and Pizulo being his students and them kind of sharing this milestone in their own tradition of modeling. Thank you Benjamin. Yes, good book and thought provoke. And this part here is just that that that tantalizing overlap of obvious and arcane. I mean, what could be more obvious in every day than every every look, every breath you take? all of that. But then it's

like but then the cognitive and active inference gets transposed into this other space that's like well no we would need to have some we need to do something different to see how it applies day-to-day versus just grounding it in our basic physiological action and then using that to kind of be like well was there an implicit reward reward oriented model? Like, did you believe that you were moving your eyes or choosing restaurants or deciding what to do around the house because of reward? Or was it just something that hadn't been considered? >> Yeah, Benjamin, go for it. >> So, tell me if this makes any sense. Yeah, earlier one of the clauses you pointed out was a little potent like controversial maybe was that active inference provides like a complete model of cognition or something like I mean it was something more qualified than that but similar but it seems like yeah that's only sort of true because it's only as complete as your mo yeah only as complete as your like sensors and like your ability to actually like record the data and information you're getting from your environment. And so the there you know the equate the model the you can model the dynamics but it's not this active inference doesn't does isn't complete because it doesn't have a complete description of exactly what to measure how to record it how to merge new data with and model your priors and it's not fully complete unless the model designer suggests you know comes up with those if that makes like does that make any sense that my argument is that it doesn't seem that complete unless >> there's a lot more description of exact actly what you know to pay attention to what and what how to model your prior and merge new updates into your prior >> exactly >> it's like what I have in this building is a complete solution to all your travel needs and it's like oh it's a car it's like no it's just metal and it's like but that's not a solution to what I needed but that's the what that was the solution to what you needed Michael Hi, I just um I wanted to sort of a side note but in terms of getting more detail on the implementation and things like that um is uh Sanjie's book that's coming in March do you think Daniel that that will be um building building out that framework a little bit more? Oh yeah, he's he's a fantastic developer and that book will have a huge number of notebooks and scripts. The math is more comprehensive and it's going to be completely tied to open source code in a much stronger way. We're going to pivot our textbook group focus there. >> Awesome. Yeah. Okay. Thank you all. Till next time.